

Education

The University of Texas at Austin

Austin, Texas

Master of Science in Computer Science, GPA 3.97/4.0

- Teaching Assistant for Graduate Parallel Systems (2 Semesters): Mentored students in concurrent programming and distributed architectures. Engineered a high-performance, GPU-accelerated K-Means clustering algorithm in **CUDA C++**, explicitly optimizing memory hierarchy and parallel thread execution for heterogeneous compute.
- Specialized Coursework: Quantum Computing (under Prof. Scott Aaronson). Developed a rigorous foundation in Fault-Tolerant Quantum Computing (FTQC), Quantum Error Correction (QEC) including Shor's 9-qubit code, and algorithmic state spaces.

National Taiwan University

Taipei, Taiwan

Master of Science in Computer Science, GPA 4.25/4.3

- Thesis "Alleviating Deduplication-oriented Fragmentation of SSDs by Jointly Considering File Hotness and Popularity" published in The 20th IEEE International SoC Conference, advised by Prof. Tei-Wei Kuo and Prof. Chien-Chung Ho
- Engineered hardware optimization algorithms in C++ by extending MQSim, a large-scale C++ hardware simulator, successfully modeling low-level data flow, real-time system performance, and architecture behavior.

National Taiwan University

Taipei, Taiwan

Bachelor of Science in Physics, minor in Computer Science, GPA 3.99/4.3

- Dual focus provides a strong mathematical foundation for understanding quantum hardware physics, analog control systems, and computational limits.
- Recipient of the Presidential Award (top 5% of class, 3 semesters) and Teaching Assistant for Numerical Analysis.

Work Experience

Oracle

Redwood City, CA

Software Engineer

November 2024 – Present

- Engineered extensible integration toolchains (via the MCP protocol) that allow third-party AI models to interface dynamically with enterprise data systems, directly aligning with requirements to productize AI infrastructure.
- Architected and implemented an LRU caching layer with TTL for module resolution, accelerating script execution performance by 3.1x and resolving critical real-time latency bottlenecks.
- Profiled and optimized Just-In-Time (JIT) compilation performance in collaboration with the GraalVM core compiler team, executing deep analysis of JFR dumps to drive significant improvements in runtime efficiency.
- Prototyped a hybrid execution engine during a corporate hackathon that integrated LLMs and vector search to parse and execute unstructured queries, demonstrating the ability to bridge novel compute frameworks with classical backends.

Oracle

Austin, Texas

Software Engineer Internship

January 2022 – August 2022

- Engineered scalable backend web services in Java and TypeScript, optimizing complex relational database queries to enhance data retrieval and heterogeneous system throughput.
- Managed continuous integration and delivery (CI/CD) pipelines via TeamCity, systematically triaging and resolving complex build integrations to stabilize deployment workflows and ensure high product quality.

An-Hui Information Technology Co., Ltd

Taipei, Taiwan

Full Stack Developer

July 2021 – December 2021

- Engineered high-concurrency backend services in Go, designing scalable architectures to manage complex transaction logic for distributed workloads.
- Architected containerized observability environments using Docker, integrating Grafana and Prometheus for real-time monitoring of distributed system health and metrics.

Technical Skills

Languages: C++, CUDA C, Java, Go, Python, JavaScript/TypeScript, Bash, SQL

Systems & Architecture: Compiler Optimization (GraalVM, AST, IR), Distributed Systems, Hardware Simulation

Technologies & Infrastructure: Linux, Docker, CI/CD (TeamCity, GitLab), Perforce, PyTorch, HuggingFace